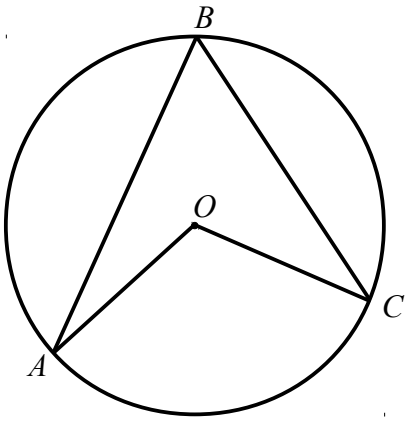
**1**

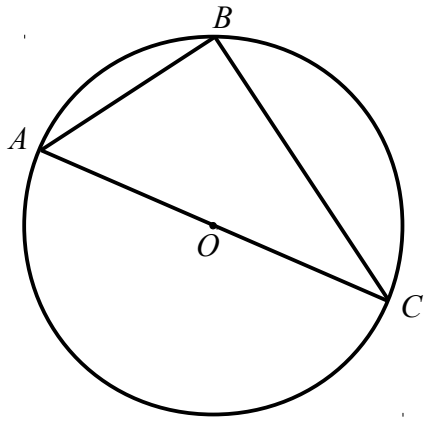



$A$, $B$ and $C$ are points on the circumference of a circle, centre $O$.

Prove that angle $AOC$ is twice the size of angle $ABC$.
You must **not** use any circle theorems in your proof.

**(Total for Question 1 is 4 marks)**

**2**

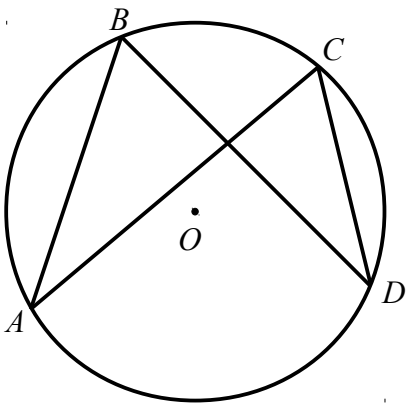



*A*, *B* and *C* are points on the circumference of a circle, centre *O*.
*AOC* is a diameter of the circle.

Prove that angle *ABC* is 90°
You must **not** use any circle theorems in your proof.

**(Total for Question 2 is 4 marks)**

**3**

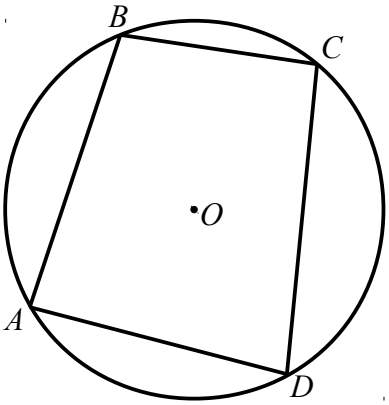



*A*, *B, C* and *D* are points on the circumference of a circle, centre *O*.

Prove that angle *ABD* and angle *ACD* are equal.

**(Total for Question 3 is 2 marks)**

**4**

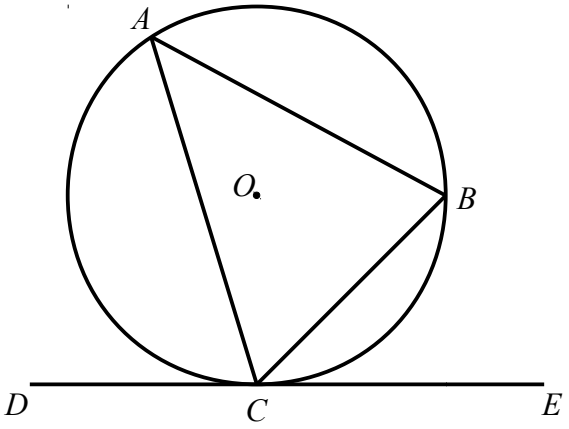



*A*, *B, C* and *D* are points on the circumference of a circle, centre *O*.

Prove that angle *ABC* and angle *ADC* add to 180°

**(Total for Question 4 is 4 marks)**

**5**




*A*, *B* and *C* are points on the circumference of a circle, centre *O*.
*DCE* is a tangent to the circle.

Prove that angle *BCE* and angle *BAC* are equal.

**(Total for Question 5 is 4 marks)**